Accelerated C# Learning Roadmap

12-Week Intensive Master Plan • October 9, 2026 – December 31, 2026

TOTAL DURATION	WEEKLY COMMITMENT	PRIMARY TARGET
12 Weeks (84 Days)	**18 – 22 Hours / Week**	**ASP.NET Core Web API Portfolio**

Phase 0 & 1: Setup & C# Fundamentals

Weeks 1–2 • Oct 9 – Oct 22

Goal: Master CLI workflows, control flow, methods, and collections.

Week 1 (Oct 9 – Oct 15): Core Syntax & Control Flow

- **Topics:** .NET SDK setup, dotnet CLI (`new`, `build`, `run`), primitive types, `for`/`foreach`/`while` loops, `if`/`else`, switch expressions.

Hands-on Project: Calculator with menu & interactive text-based quiz game.

Week 2 (Oct 16 – Oct 22): Methods, Collections & Debugging

- **Topics:** Methods, array/`List<T>` manipulation, basic exception handling (`try`/`catch`), nullability operators, debugger stepping & breakpoints.

Hands-on Project: Console To-Do List application committed & structured on GitHub.

Phase 2: Object-Oriented Programming

Weeks 3–4 • Oct 23 – Nov 5

Goal: Model real-world domains cleanly using OOP design principles.

Week 3 (Oct 23 – Oct 29): Encapsulation & Abstraction

- **Topics:** Classes, objects, constructors, properties, access modifiers (`private`, `public`, `internal`), static members, namespaces.

Hands-on Project: Bank Account System (accounts, transactions, validation, error handling).

Week 4 (Oct 30 – Nov 5): Inheritance, Polymorphism & Interfaces

- **Topics:** Interfaces vs. Abstract Classes, overriding (`virtual`/`override`), records, value vs. reference types.

Hands-on Project: Inventory or Library Management System utilizing interface abstractions.

Phase 3: Intermediate C# & Data Handling

Weeks 5–6 • Nov 6 – Nov 19

Goal: Write idiomatic C# with LINQ, collections, and unit testing.

Week 5 (Nov 6 – Nov 12): Collections, LINQ & File I/O

- **Topics:** `Dictionary`, `HashSet`, LINQ operations (`Where`, `Select`, `GroupBy`, `OrderBy`), `System.Text.Json` serialization.

Hands-on Project: JSON Expense Tracker with LINQ automated report generation.

Week 6 (Nov 13 – Nov 19): Delegates, Lambdas & Unit Testing

- **Topics:** `Func`/`Action`, Extension Methods, Unit Testing with xUnit using the Arrange-Act-Assert pattern.

Hands-on Project: Add a full xUnit test suite to your Expense Tracker or Bank System.

Phase 4: Advanced C# & Runtime Concepts

Weeks 7–8 • Nov 20 – Dec 3

Goal: Master asynchronous code execution, memory mechanics, and dependency injection.

Week 7 (Nov 20 – Nov 26): Asynchronous Programming & Memory

- **Topics:** `async/await`, `Task`, `CancellationToken`, `IDisposable`, `using` statements, Stack vs. Heap allocation.

Hands-on Project: Async Web/Data Downloader with progress tracking and cancellation support.

Week 8 (Nov 27 – Dec 3): Dependency Injection & SOLID

- **Topics:** Dependency Injection (DI), SOLID principles, core patterns (Repository, Factory, Strategy).

Hands-on Project: Refactor a prior console application to implement DI and SOLID architecture.

Phase 5: Specialization – Web API & Databases

Weeks 9–11 • Dec 4 – Dec 24

Goal: Build and deploy a production-ready Web API backend.

Week 9 (Dec 4 – Dec 10): ASP.NET Core Fundamentals

- **Topics:** RESTful API design, Controllers vs. Minimal APIs, Routing, Middleware pipeline.

Week 10 (Dec 11 – Dec 17): Databases & Entity Framework Core

- **Topics:** Entity Framework Core (EF Core), DbContext, Migrations, CRUD operations, relational querying.

Week 11 (Dec 18 – Dec 24): Authentication & Deployment

- **Topics:** JWT Authentication & Authorization, OpenAPI/Swagger documentation, Docker basics.

Hands-on Project: Build a complete REST API backend (e.g., E-Commerce/Task Management) with EF Core, JWT Auth, and Swagger.

Phase 6: Portfolio Polish & Delivery

Week 12 • Dec 25 – Dec 31

Goal: Prepare repository for technical review and public presentation.

Week 12 (Dec 25 – Dec 31): Integration, Testing & Documentation

- **Topics:** Integration testing, basic performance tuning, structured logging, writing exemplary GitHub READMEs.

Deliverable: Publish full Web API project on GitHub with architectural documentation, tests, and deployment setup.

Execution Rules for Success

1. Time Allocation

Target 2–3 hours on weekdays and 4–5 hours on weekends. Consistency creates momentum.

2. 70/30 Practice Ratio

Spend 30% studying concepts and 70% actively writing code and debugging errors.

3. Daily Version Control

Commit your progress daily to GitHub to showcase continuous proof of work.